MULTIMODAL INTERACTIVE TECHNOLOGIES AND APPLICATIONS

Project Concept

Botanical Garden

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1 Vision

MITA is a browser-based botanical garden combining **WebVR panoramic exploration** with a **conversational, voice-driven assistant**. Users walk through 360° panoramic scenes, point at 3D plant models and speak naturally, turning a static exhibit into a fluid multimodal dialogue.

2 Characterizing Features

- **Immersive 3D environment.** A-Frame (WebVR) renders five linked panoramic scenes with clickable portals and inspectable 3D plant models.
- **Multimodal input.** Gaze/pointer raycasting, speech (STT via the Web Speech API), and click/touch fallback.
- **Multimodal output.** Spatial highlight on the selected plant, adaptive 2D HUD cards (botanical, medicinal, comparison, fallback), and TTS narration.
- **LLM-driven dialogue.** A stateless backend performs intent parsing and slot extraction (medicinal, botanical, toxicity, drought, feng shui) with local fallback when the LLM is unavailable.
- **Client-side state.** Selected plant, visited plant history, ambiguity candidates and dialogue history are tracked client-side, enabling multi-turn reasoning without a server session.
- **Robust disambiguation.** Overlapping candidates are ranked by depth, labelled A / B / C, and resolved by voice or overlay click, with two retry attempts before pointer fallback.

3 Target Users and Why It Is Multimodal

The system targets **casual museum visitors, botany students and accessibility-oriented users** on desktop, tablet or VR headset. MITA is multimodal **by design**: every interaction fuses pointing and speech (input) with 3D highlight, visual card and TTS (output), and degrades gracefully when any single modality fails.